DATA STRUCTURE

LAB PROGRAMS

12. a) Queue Insertion (Enqueue) Using Array

```
#include <stdio.h>
#define MAX 10
int main()
{
   int queue[MAX];
   int front = 0, rear = -1;
   int n, i, value;
   printf("Enter number of elements to insert: ");
   scanf("%d", &n);
   for(i = 0; i < n; i++)
   {
      if(rear == MAX - 1)
      {
         printf("Queue Overflow\n");
         break;
      }
      printf("Enter element: ");
      scanf("%d", &value);
      rear++;
      queue[rear] = value;
   }
   printf("Queue elements are: ");
   for(i = front; i <= rear; i++)
```

```
    {
        printf("%d ", queue[i]);
    }
    return 0;
}
```

Output

```
Enter number of elements to insert: 4
Enter element: 36
Enter element: 58
Enter element: 69
Enter element: 79
Queue elements are: 36 58 69 79

=== Code Execution Successful ===
```

b) Queue Deletion (Dequeue) Using Array

```
#include <stdio.h>
#define MAX 10
int main()
{
    int queue[MAX];
    int front = 0, rear, n, i;
    printf("Enter number of elements in queue: ");
    scanf("%d", &n);
```

```
    rear = n - 1;
    printf("Enter queue elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &queue[i]);
    }
    if(front > rear)
    {
        printf("Queue Underflow");
    }
    else
    {
        printf("Deleted element = %d\n", queue[front]);
        front++;
        printf("Queue after deletion: ");
        for(i = front; i <= rear; i++)
        {
            printf("%d ", queue[i]);
        }
    }
    return 0;
}
```

Output

```
Enter number of elements in queue: 5
Enter queue elements:
12 32 54 65 67
Deleted element = 12
Queue after deletion: 32 54 65 67

=== Code Execution Successful ===
```

c) Queue Display Using Array

```
#include <stdio.h>
#define MAX 10
int main()
{
    int queue[MAX];
    int front = 0, rear, n, i;
    printf("Enter number of elements in queue: ");
    scanf("%d", &n);
    rear = n - 1;
    printf("Enter queue elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &queue[i]);
    }
    if(front > rear)
```

```
    {
        printf("Queue is Empty");
    }
    else
    {
        printf("Queue elements are: ");
        for(i = front; i <= rear; i++)
        {
            printf("%d ", queue[i]);
        }
    }
    return 0;
}
```

```
Output
Enter number of elements in queue: 3
Enter queue elements:
34 46 67
Queue elements are: 34 46 67

=== Code Execution Successful ===
```